

EventSphere - Master Project Blueprint

React + Spring Boot + Spring AI Portfolio Project

1. Vision

Build a production-ready event management and networking platform for hackathons, workshops and conferences. It should demonstrate authentication, REST APIs, WebSockets, AI integration, clean architecture and deployment.

2. Functional Modules

Authentication

- JWT Login
- Google OAuth
- Forgot Password
- Email Verification

Organizer

- Create/Edit/Delete Events
- Publish Events
- QR Ticket Generation
- Attendance Dashboard

Participant

- Browse Events
- Register
- Personal Dashboard
- Certificates

Networking

- Search Participants
- Connection Requests
- Real-time Chat using WebSockets

3. AI Features

- Generate event descriptions
- Improve event titles
- Create schedules automatically
- Summarize event notes
- Recommend events based on interests

4. Database Design

Users(id,name,email,password,role)
Events(id,title,description,date,location,capacity)
Registrations(id,user_id,event_id,status)
Connections(id,sender,receiver,status)
Messages(id,sender,receiver,message,time)
Attendance(id,event_id,user_id,checked_in)
Certificates(id,user_id,event_id,file_url)

5. REST APIs

POST /auth/login
POST /auth/register
GET /events
POST /events
PUT /events/{id}
DELETE /events/{id}
POST /events/{id}/register
GET /users/profile
POST /connections/request
GET /chat/messages

6. Frontend Structure

src/
■■■ components/
■■■ pages/
■■■ hooks/
■■■ services/
■■■ layouts/
■■■ store/
■■■ routes/
■■■ utils/

7. Backend Structure

controller/
service/
repository/
entity/
dto/
security/
config/
exception/
ai/
websocket/

8. UI Screens

Landing Page
Login/Register
Event Listing
Event Details
Organizer Dashboard
Participant Dashboard
Chat Screen
Analytics Dashboard
Admin Panel

9. Development Plan

Week 1: Authentication + Database + UI setup
Week 2: Event CRUD + Registration
Week 3: QR Tickets + Dashboards
Week 4: WebSockets Chat + Notifications

Week 5: Spring AI Features

Week 6: Docker + Deployment + Testing

10. Stretch Goals

Redis caching

Cloudinary uploads

GitHub Actions CI/CD

Docker Compose

Kubernetes deployment

Microservice migration